

# *Lecture 2: Regular curves, arclength, and curvature*

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**Definition.** A *parametrized curve* in  $\mathbb{R}^n$  is a smooth map

$$c : I \longrightarrow \mathbb{R}^n,$$

where  $I \subset \mathbb{R}$  is an open interval.

The parameter  $t$  records how the curve is traversed. The image

$$c(I) \subset \mathbb{R}^n$$

records only the points through which it passes.

Different parametrized curves can have the same image. For example,

$$c(t) = (\cos t, \sin t) \quad \text{and} \quad \widetilde{c}(t) = (\cos 2t, \sin 2t)$$

both trace the unit circle, but at different speeds.

The *velocity* of  $c$  at  $t$  is

$$\dot{c}(t) \in \mathbb{R}^n.$$

Its speed is

$$v(t) = \|\dot{c}(t)\|.$$

**Definition.** The curve  $c$  is regular if

$$\dot{c}(t) \neq 0$$

for every  $t \in I$ .

Regularity means that the curve has a well-defined direction of motion at every time. It does not require the curve to be injective.

For example, the figure-eight curve

$$c(t) = (\sin t, \sin 2t)$$

is regular but not injective.

**Definition.** Let  $h: J \rightarrow I$  be a smooth diffeomorphism between intervals. The curve

$$\tilde{c} = c \circ h: J \longrightarrow \mathbb{R}^n$$

is called a **reparametrization** of  $c$ .

The chain rule gives

$$\dot{\tilde{c}}(u) = \dot{c}(h(u)) \frac{dh}{du}.$$

Consequently,

$$\|\dot{\tilde{c}}(u)\| = \|\dot{c}(h(u))\| \left| \frac{dh}{du} \right|.$$

Since a diffeomorphism of intervals has  $h'(u) \neq 0$ , a reparametrization of a regular curve is again regular.

If  $h' > 0$ , the orientation is preserved. If  $h' < 0$ , the orientation is reversed.

**Definition.** The *length* of a smooth curve  $c: [a, b] \rightarrow \mathbb{R}^n$  is

$$L(c) = \int_a^b \|\dot{c}(t)\| \, dt.$$

Length measures the curve itself, not the particular way in which it is parametrized.

*Sketch of proof.* Suppose  $\tilde{c} = c \circ h$ , where  $h: [\alpha, \beta] \rightarrow [a, b]$  preserves orientation. Then

$$L(\tilde{c}) = \int_{\alpha}^{\beta} \|\dot{c}(h(u))\| h'(u) \, du.$$

The substitution  $t = h(u)$  gives

$$L(\tilde{c}) = \int_a^b \|\dot{c}(t)\| \, dt = L(c).$$

The orientation-reversing case follows in the same way using  $|h'|$ .

Let  $c : I \rightarrow \mathbb{R}^n$  be regular and fix  $t_0 \in I$ . Define the arclength function

$$s(t) = \int_{t_0}^t \|c'(u)\| du.$$

By the fundamental theorem of calculus,

$$s'(t) = \|\dot{c}(t)\| > 0.$$

Thus  $s$  is strictly increasing and, locally, has a smooth inverse

$$t = t(s).$$

The number  $s(t)$  is the signed distance traveled along the curve from the point  $c(t_0)$  to the point  $c(t)$ .

Regularity is essential because it gives  $s'(t) \neq 0$ .

## Arclength parametrization.

Every regular curve can be reparametrized locally so that it has unit speed.

*Sketch of proof.* Let  $s = s(t)$  be the arclength function and let  $t = t(s)$  be its local inverse. Define

$$\bar{c}(s) = c(t(s)).$$

Then

$$\bar{c}'(s) = c'(t(s)) \frac{dt}{ds}.$$

Since

$$\frac{ds}{dt} = \|c'(t)\|, \quad \frac{dt}{ds} = \frac{1}{\|c'(t)\|},$$

we obtain

$$\|\bar{c}'(s)\| = 1.$$

When a curve  $c$  is parameterized by arclength then we denote the unit tangent vector by:

$$c'(s) = \dot{c}(s), \quad \|c'(s)\| = 1. \quad (1)$$

Thus, if  $c$  is *any* regular curve we have

$$\dot{c} = \frac{ds}{dt} c' = \|\dot{c}\| c' \quad (2)$$

Suppose from now on that  $c = c(s)$  is parametrized by arclength, so that

$$\|c'(s)\| = 1.$$

**Definition.** The *unit tangent vector* of a curve  $c$  is

$$T(s) = c'(s).$$

Since  $\langle T, T \rangle = 1$ , differentiation gives

$$0 = \frac{d}{ds} \langle T, T \rangle = 2 \langle T', T \rangle.$$

Therefore

$$T'(s) \perp T(s).$$

The tangent vector can change only in a direction perpendicular to itself.

**Definition.** The *curvature* of a unit-speed curve  $c$  is

$$\kappa(s) = \|T'(s)\|.$$

Since  $T = c'$ , we may also write

$$\kappa(s) = \|c''(s)\|.$$

Curvature measures the rate at which the direction of the curve changes per unit distance traveled.

A straight line has constant tangent vector and therefore

$$\kappa = 0.$$

Large curvature means that the direction changes rapidly. Small curvature means that the curve is nearly straight.

**Definition.** Suppose  $c$  is a unit-speed curve and  $\kappa(s) > 0$  for all  $s$ . The *principal normal vector* is

$$N(s) = \frac{T'(s)}{\|T'(s)\|}.$$

Thus

$$T'(s) = \kappa(s)N(s).$$

The vector  $N$  points toward the direction in which the curve is turning.

The number

$$\rho(s) = \frac{1}{\kappa(s)}$$

is called the radius of curvature. The corresponding center of curvature is

$$c(s) + \rho(s)N(s).$$

When  $\kappa = 0$ , the principal normal is not defined.

Curvature is geometric and should not depend on the speed of a chosen parameter. For a regular curve  $c = c(t)$ , set

$$T(t) = \frac{c'(t)}{\|c'(t)\|}.$$

Since

$$\frac{d}{ds} = \frac{1}{\|c'(t)\|} \frac{d}{dt},$$

the curvature is

$$\kappa(t) = \left\| \frac{dT}{ds} \right\| = \frac{1}{\|c'(t)\|} \left\| \frac{dT}{dt} \right\|.$$

For a curve in  $\mathbb{R}^3$ , write

$$v(t) = \|c'(t)\|.$$

Since  $\|T(t)\| = 1$ , differentiation gives

$$\left\langle T, \frac{dT}{dt} \right\rangle = 0.$$

Thus  $T$  and  $dT/dt$  are perpendicular, and hence

$$\left\| \frac{dT}{dt} \right\| = \left\| T \times \frac{dT}{dt} \right\|.$$

Now differentiate  $T = c'/v$ :

$$\frac{dT}{dt} = \frac{c''}{v} - \frac{v'}{v^2}c'.$$

Taking the cross product with  $T = c'/v$ , the second term disappears:

$$T \times \frac{dT}{dt} = \frac{c'}{v} \times \left( \frac{c''}{v} - \frac{v'}{v^2}c' \right) = \frac{c' \times c''}{v^2}.$$

Therefore

$$\kappa(t) = \frac{1}{v} \left\| \frac{dT}{dt} \right\| = \frac{\|c'(t) \times c''(t)\|}{v^3} = \frac{\|c'(t) \times c''(t)\|}{\|c'(t)\|^3}.$$

This formula allows us to compute curvature without first finding an arclength parametrization.

### Example

Consider a line

$$c(t) = p + tv, \quad v \neq 0.$$

Then

$$c'(t) = v, \quad c''(t) = 0.$$

Hence

$$\kappa(t) = 0.$$

This matches the geometric interpretation: the direction of the line never changes.

Conversely, a unit-speed curve satisfying  $\kappa = 0$  on an interval has

$$T' = 0.$$

Thus  $T$  is constant, and integration gives

$$c(s) = p + sT.$$

Therefore a unit-speed curve has zero curvature precisely when it is a line.

### Example

Consider the circle of radius  $R$ , parametrized by arclength:

$$c(s) = \left( R \cos \frac{s}{R}, R \sin \frac{s}{R} \right).$$

Then

$$T(s) = c'(s) = \left( -\sin \frac{s}{R}, \cos \frac{s}{R} \right)$$

and

$$T'(s) = \frac{1}{R} \left( -\cos \frac{s}{R}, -\sin \frac{s}{R} \right).$$

Consequently,

$$\kappa = \|T'(s)\| = \frac{1}{R}.$$

Small circles bend more sharply than large circles.

### Example

Consider the parabola

$$c(t) = (t, t^2).$$

Then

$$c'(t) = (1, 2t), \quad c''(t) = (0, 2).$$

Using the curvature formula gives

$$\kappa(t) = \frac{2}{(1 + 4t^2)^{3/2}}.$$

The curvature is largest at the vertex:

$$\kappa(0) = 2.$$

It approaches zero as  $|t| \rightarrow \infty$ , reflecting the fact that the parabola becomes progressively flatter.

The parameter  $t$  is not arclength, but the curvature formula still produces the geometric curvature.

For a unit-speed curve, Taylor expansion gives

$$c(s+h) = c(s) + hT(s) + \frac{h^2}{2}\kappa(s)N(s) + O(h^3).$$

The first-order term

$$c(s) + hT(s)$$

describes the tangent line.

The second-order term

$$\frac{h^2}{2}\kappa(s)N(s)$$

records the first bending away from that line.

When  $\kappa(s) > 0$ , the circle with radius  $1/\kappa(s)$ , tangent direction  $T(s)$ , and center in the direction  $N(s)$  is called the osculating circle. It is the circle that best approximates the curve to second order at  $c(s)$ .